

8th Progress Report

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Date:	19/03/2024
Subject:	RL for Flying-V
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Project term:	01/2024 - 08/2024
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Outcome of the previous meeting

- Discussed how more model-matching tests can be performed.
- Presented decision to use short-period for preliminary research.
- Presented update on the implementation of IDHP.
- Discussed distinction between eligibility trace and momentum gradient descent, RLS covariance windup.
- Got interesting papers on eligibility vs momentum.

Description of the results since last meeting

- Properly understood the distinction between eligibility trace and momentum: eligibility trace only uses the latest TD error for parameter update, while momentum uses all previous TD errors for update (which biases updates towards previous TD errors). For explanation and sources, see thesis notes¹, but Nichols [1] summarizes it very well.
- Updated Linearization to exclude actuator dynamics, used linmod instead of linearization for easier linearization point specification. Performed additional model matching tests for 3 models, the nonlinear dynamics with all states, the linearized dynamics with all states, and linearized dynamics with only short period states, see Figure 1. Also, realized that the sign convention for elevator deflection does not use the normal body reference frame (xyz = nose, right wing, belly), in the models the airplane pitches up when elevator deflection is positive.
- Wrote down the entire set of update rules for IDHP in a pdf, this will be useful for implementing IDHP and for debugging, still need to flesh out this pdf more by e.g. defining/checking the dimensions of all variables, specifying the order of update rules... and possibly more (I'm open to suggestions).
- Attended Flying-V progress meeting where I learned about the IP agreement. I emailed Roelof asking for clarification on the extent of the agreement. Still considering if I should move forward with using the Flying-V, as a backup I have submitted a new knowledge embargo form and got the short-period model of the citation as well.

¹<https://shorturl.at/dlmuI>, the [[@L ...]] strings are links to other files, this was intended to be viewed in obsidian/some markdown viewer.

- Implemented a working version of IDHP, had many hidden bugs due to not using `.copy()` for arrays, realized there are a lot of hyperparameters that I can theoretically tune (most important ones so far: adaptive learning rate, network weight initialization, network sizes, network inputs), see examples in Figure 2. Did a number of hyperparameter runs to find the parameters giving the best tracking performance. 19th of March marks the last day planned for implementing IDHP in my Gantt chart, so I am on schedule.

Plans for next 2 weeks

1. Clean up current code:
 - Add doc strings
 - Redo hard coded variables
 - Add assertion checks
 - Revisit folder and file structures...
2. Do some more experiments with vanilla IDHP:
 - Observe effects of varying network sizes
 - Try annealed learning rates instead of binary high/low rates.
 - Maybe try setting up multiprocessing scripts for faster data gathering.
3. Begin experiments with eligibility traces. Comparison between vanilla and eligibility IDHP shall use the same hyperparameters when shared (the optimal set of parameters for vanilla IDHP)!
4. Do some hyperparameter runs on Citation model for backup, and revisit research objectives to see where they can be tweaked for using Citation as the research platform (ideas: add more emphasize on researching IDHP algorithms).

References

- [1] Barry D. Nichols. “A comparison of eligibility trace and momentum on SARSA in continuous state-and action-space”. In: *2017 9th Computer Science and Electronic Engineering (CEECE)*. 2017, pp. 55–59. DOI: 10.1109/CEECE.2017.8101599.

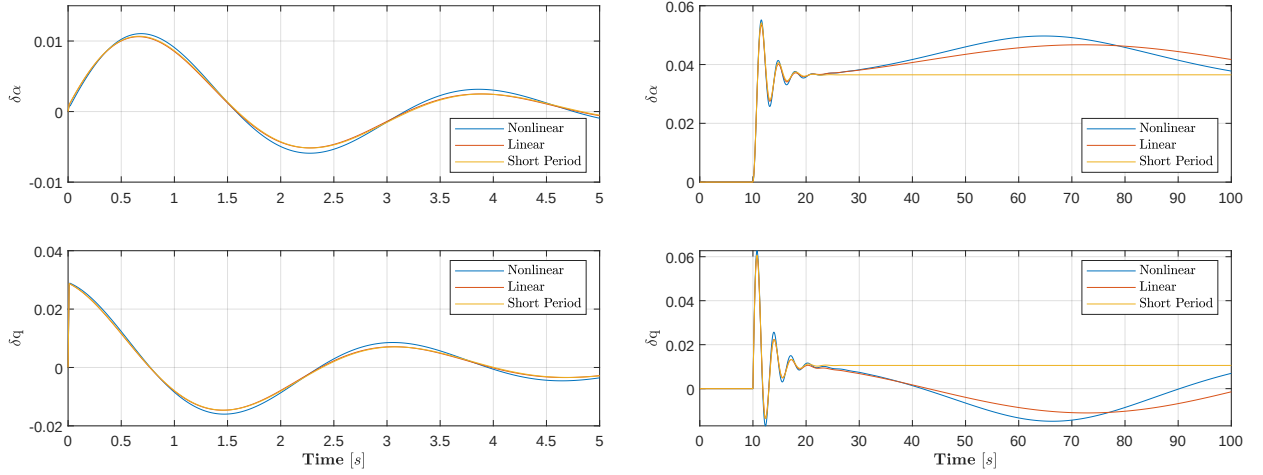
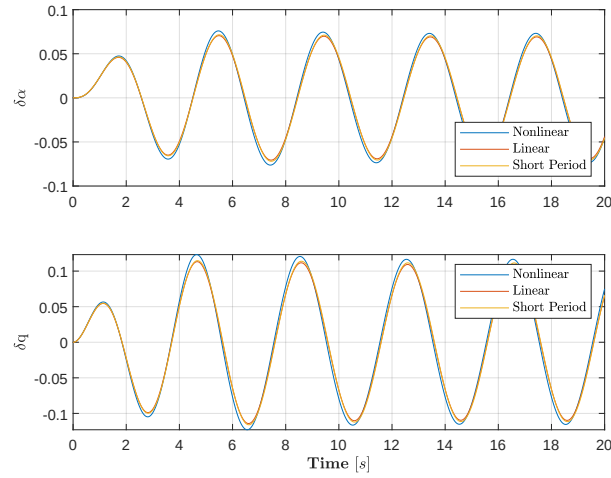
(a) $\delta_e = 100$ [deg] if timestep == 1st else 0(b) $\delta_e = 5$ [deg] if $t \geq 5$ (c) $\delta_e = 5 \sin(\frac{\pi}{2}t)$ [deg]

Figure 1: Plot of the α [deg] and q [deg/s] responses from the nonlinear, linear, and short-period models of the system.

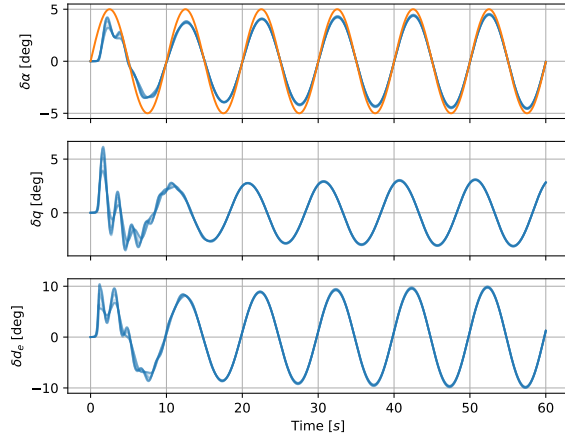
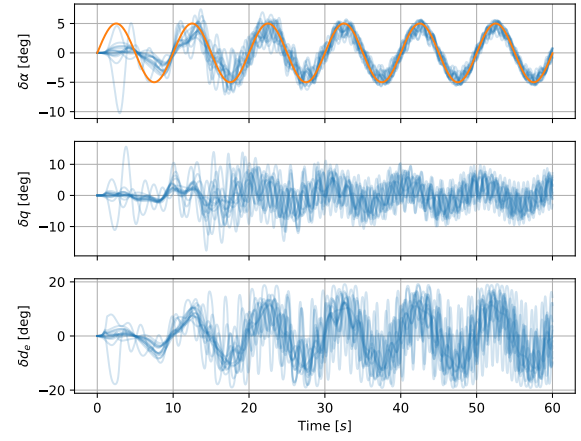
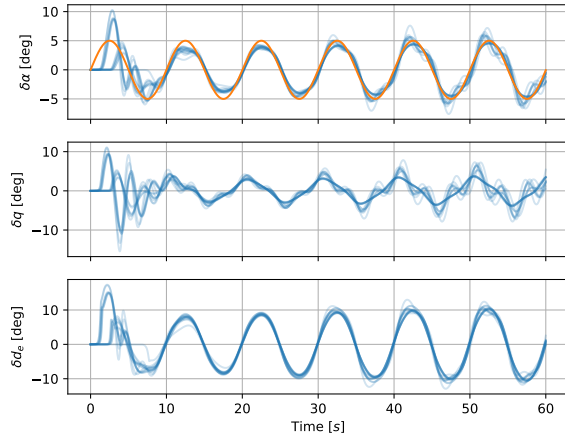
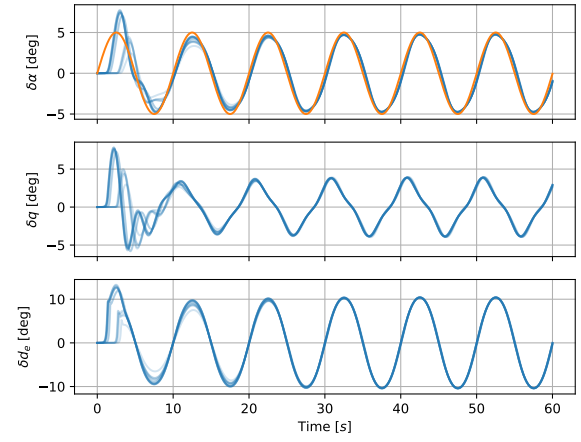
(a) Only tracking error as input, return = -7.78 .(b) Tracking error as input, return = -39.26 (c) Full aircraft states reference Aoa as network inputs, return = -21.13 .(d) Aircraft and reference Aoa as network inputs, return = -11.96 .

Figure 2: Three examples of IDHP running on AoA tracking.